TARUN KUMAR

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SUMMARY
Driven Robotics & Automation Engineering student building autonomous mobile systems utilizing ROS2, Nav2, and SLAM. Proven experience in simulation, real-world physical deployment, and computer vision. Active open-source contributor to ROS2 Navigation2. Seeking engineering internships in robotics and autonomous systems.

OPEN SOURCE CONTRIBUTIONS

ROS2 NAVIGATION2 (NAV2)

- Contributed to the core Nav2 stack by improving documentation and configuration clarity for Costmap2D parameters.
- Pull request successfully reviewed and merged by the core Nav2 maintainer, directly aiding the ROS developer community.

ENGINEERING PROJECTS

Autonomous Mobile Robot (Hardware) | *ROS2 Jazzy, Nav2, SLAM Toolbox, Python, C++, Raspberry Pi 5*

- Engineered a custom differential drive robot for real-time spatial awareness, indoor mapping, and dynamic path planning.
- Tuned global and local costmaps for robust navigation through complex environments.
- Developed custom hardware-middleware interface nodes on a Raspberry Pi 5.

Hybrid Autonomous Navigation System | *ROS2, Nav2, OpenCV, Gazebo, Twist Multiplexer, PID Control*

- Architected an industrial-grade hybrid navigation simulated AGV combining global Nav2 path planning with local OpenCV-based vision guidance.
- Developed a custom vision pipeline utilizing OpenCV (HSV filtering) and PID control for high-precision local line-tracking.
- Implemented a Twist Multiplexer to autonomously resolve command velocity conflicts between the global Nav2 controller and the local vision node.

Agridrone Audit (Computer Vision Payload) | *YOLOv5, OpenCV, Edge AI, MAVLink*

- Built an intelligent perception system for agricultural UAVs using real-time Computer Vision for Evapotranspiration sensing.
- Implemented a YOLOv5 model for crop health detection and mapped aerial vision data with precise GPS coordinates via telemetry parsing.

Autonomous UAV System (Flight Control) | *GPS Navigation, Flight Controllers, MAVLink*

- Developed a highly stable UAV platform with Team Challengers for precision autonomous missions and position-hold.
- Calibrated PIDs and configured complex autonomous failsafes including autonomous emergency landing and RTL scenarios.

ArticulateX (Robotic Arm Manipulation) | *Kinematics (FK/IK), Trajectory Planning, Embedded Systems*

- Modeled the forward and inverse kinematics for a high-versatility multi-DoF articulated robotic arm.
- Programmed low-level motion control firmware for precise multi-axis joint synchronization and path execution.

Smart Surveillance Robot | *Raspberry Pi, Arduino, Serial Comms, Distributed Control*

- Built a modular surveillance platform utilizing distributed control architecture separating high-level perception (Pi) from low-level mechanical control (Arduino) using asynchronous UART communication.

SmartRoomOccupancy (Edge ML Modeling) | *AI/ML, Data Modeling, Sensor Fusion*

- Managed the complete ML lifecycle to deploy data models on resource-constrained edge hardware for accurate occupancy detection.

Mechanical Design Portfolio | *SolidWorks, Ansys Workbench, FEA, Aerodynamics*

- Managed the complete CAD-to-Simulation workflow for a portable UAV catapult launcher and robot arm, validating aerodynamic loads and mechanical stress via FEA.

TECHNICAL EXPERTISE

- **Languages:** C++, Python, TypeScript, MATLAB
- **Robotics & Simulation:** ROS2, Nav2, SLAM Toolbox, Gazebo, URDF, Auterion / PX4
- **Computer Vision & ML:** OpenCV, YOLOv5, Edge AI Deployment, Sensor Fusion

- **Hardware & Embedded:** Raspberry Pi, Arduino, Microcontrollers, UART/Serial Comms
- **Engineering Tools:** Git, SolidWorks, Ansys, Linux (Ubuntu), MAVLink

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EDUCATION

SAHYADRI COLLEGE OF ENGINEERING & MANAGEMENT

| Mangalore, India

BE in Robotics & Automation Engineering | 2022 – 2026

National Institute of Technology Karnataka (NITK) | Surathkal, India

Research Intern (UAV Systems) | Summer 2024

CERTIFICATIONS

- **SOLIDWORKS CSWA** (CAD Design Associate)
- **MathWorks** (MATLAB Onramp & Simulink Onramp)
- **Python Foundation Certification**